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Tamil Nadu Innovation Modernization through
Problem Solving and Advanced Creative Tech



TN-IMPACT 2026

IDENTIFYING INDUSTRIAL CHALLENGES

PROBLEM STATEMENT NUMBER :TNI26061

PROBLEM STATEMENT TITLE :IoT-Based Real-Time Cargo Theft
Detection and Security System



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Team Name : ElectroMinds

College Mentor Name : Dr. S .Sheikameer Batcha

College Name: Sri Eshwar College of Engineering

Project Developed by : KISHORE C



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Problem statement

Cargo theft is a major **challenge in the logistics industry**, leading to significant **financial losses** and **supply chain disruptions**. **Traditional security** methods such as **mechanical locks** and **seals** cannot detect unauthorized access in real time. As a result, **theft** is often **discovered only after delivery**. Therefore, an **intelligent monitoring system** is required to **detect tampering** and **intrusion during transportation**.



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Problem description



Cargo containers transported by trucks are vulnerable to **theft during transit, parking, or storage in poorly monitored areas**. **Criminals** may tamper with container **locks or remove goods** without being detected. **Existing solutions** such as **GPS tracking only monitor location and cannot detect physical intrusion**. Therefore, a **smart monitoring system** is required to **detect suspicious activities and improve cargo security**.



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Objectives

- To design and develop an **IoT-based smart cargo container security system.**
- To detect **unauthorized door opening** and **lock tampering** using **sensor technology.**
- To **monitor human presence** and **suspicious activity near the container.**
- To **detect cargo displacement** or **internal movement** inside the container.
- To provide **real-time alerts** and **location tracking** for **improved cargo security.**



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Proposed Solution

The proposed system is an **IoT-based smart cargo container security system** designed using a **dual-node architecture** consisting of a **Master Node** and a **Child Node**. The **Child Node monitors** the container **door** and **lock status using a reed switch and accelerometer**. When tampering or unauthorized door opening is detected, it sends a trigger signal to the **Master Node**. The Master Node **activates sensors** such as **microwave radar, PIR sensor, and VL53L5CX time-of-flight sensor** to monitor human presence and cargo displacement. If **suspicious activity is detected**, the system **sends an alert along with location information**.

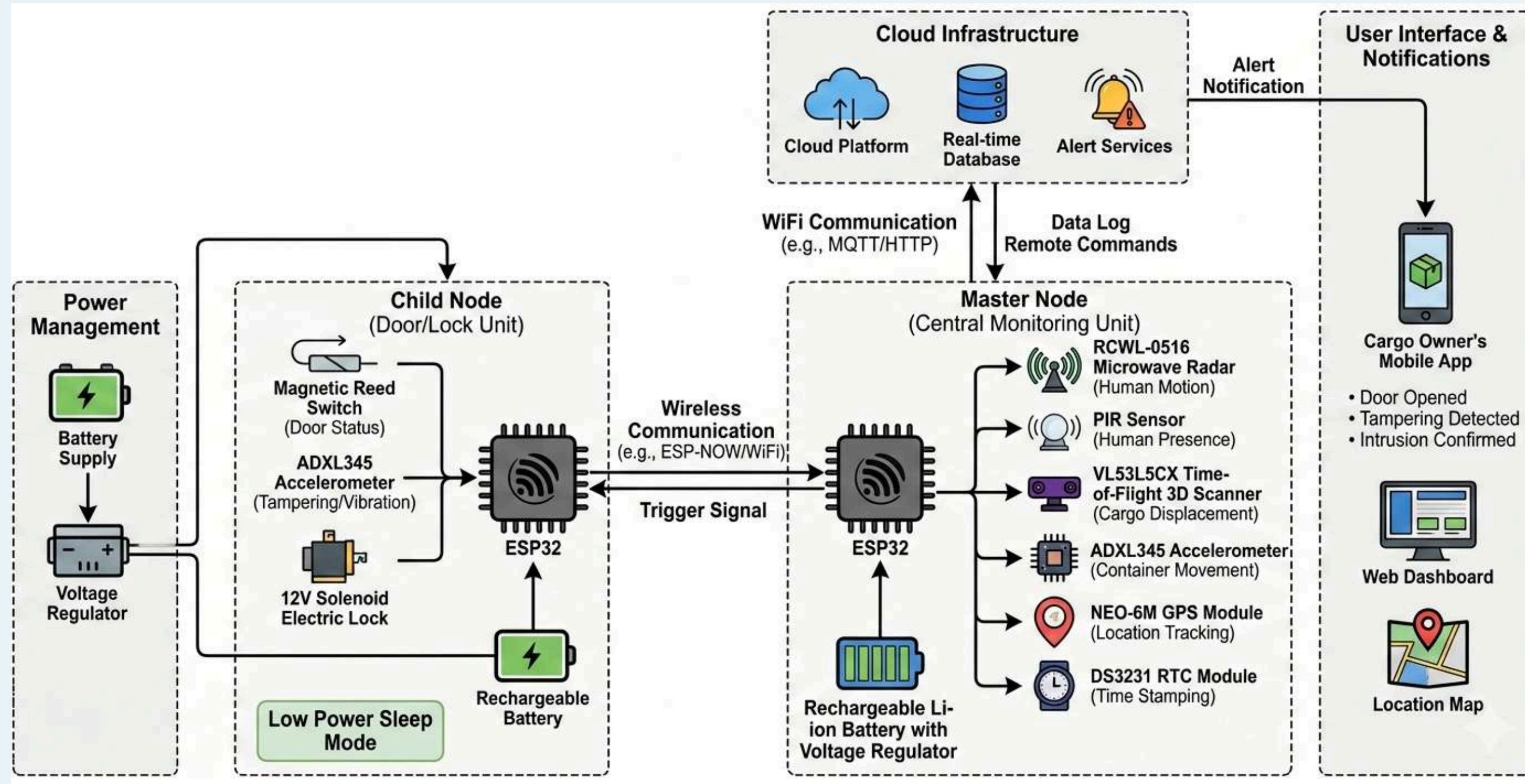


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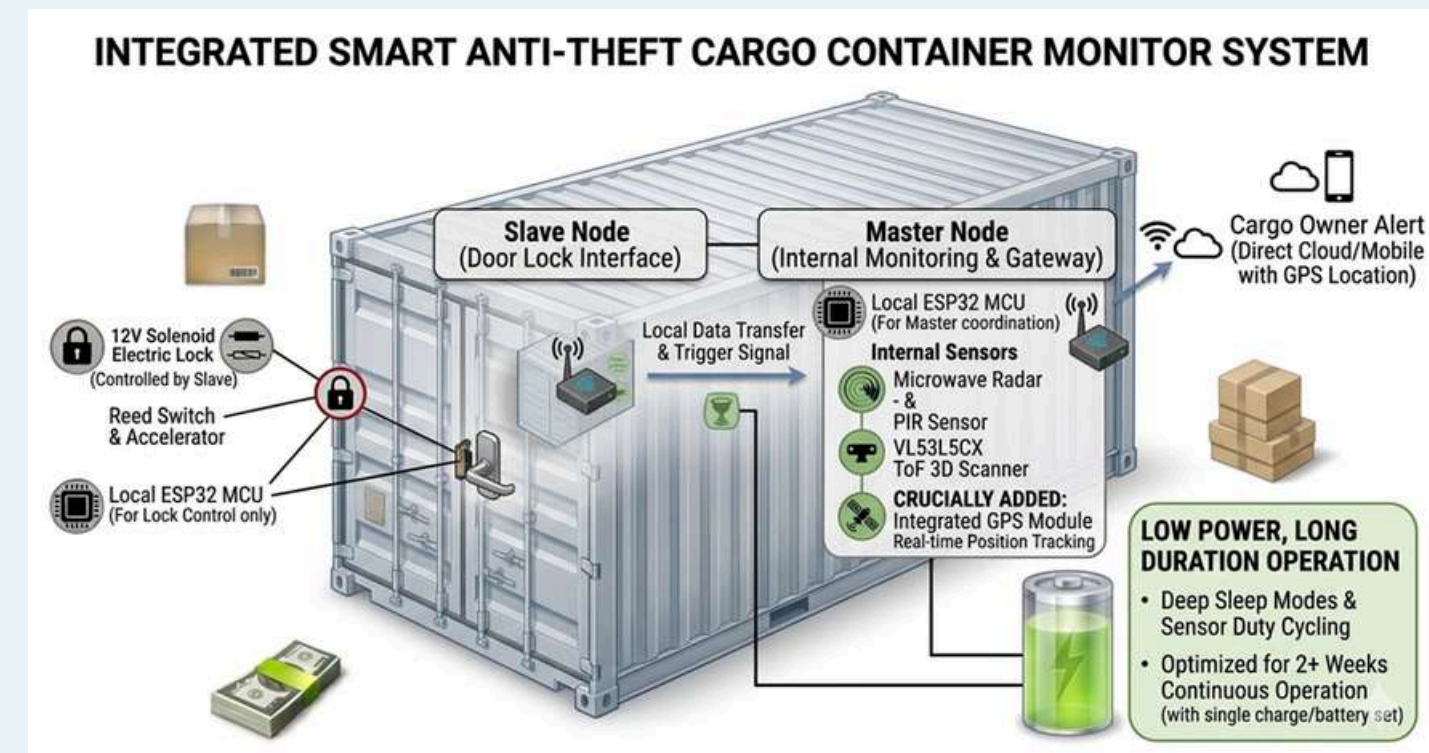
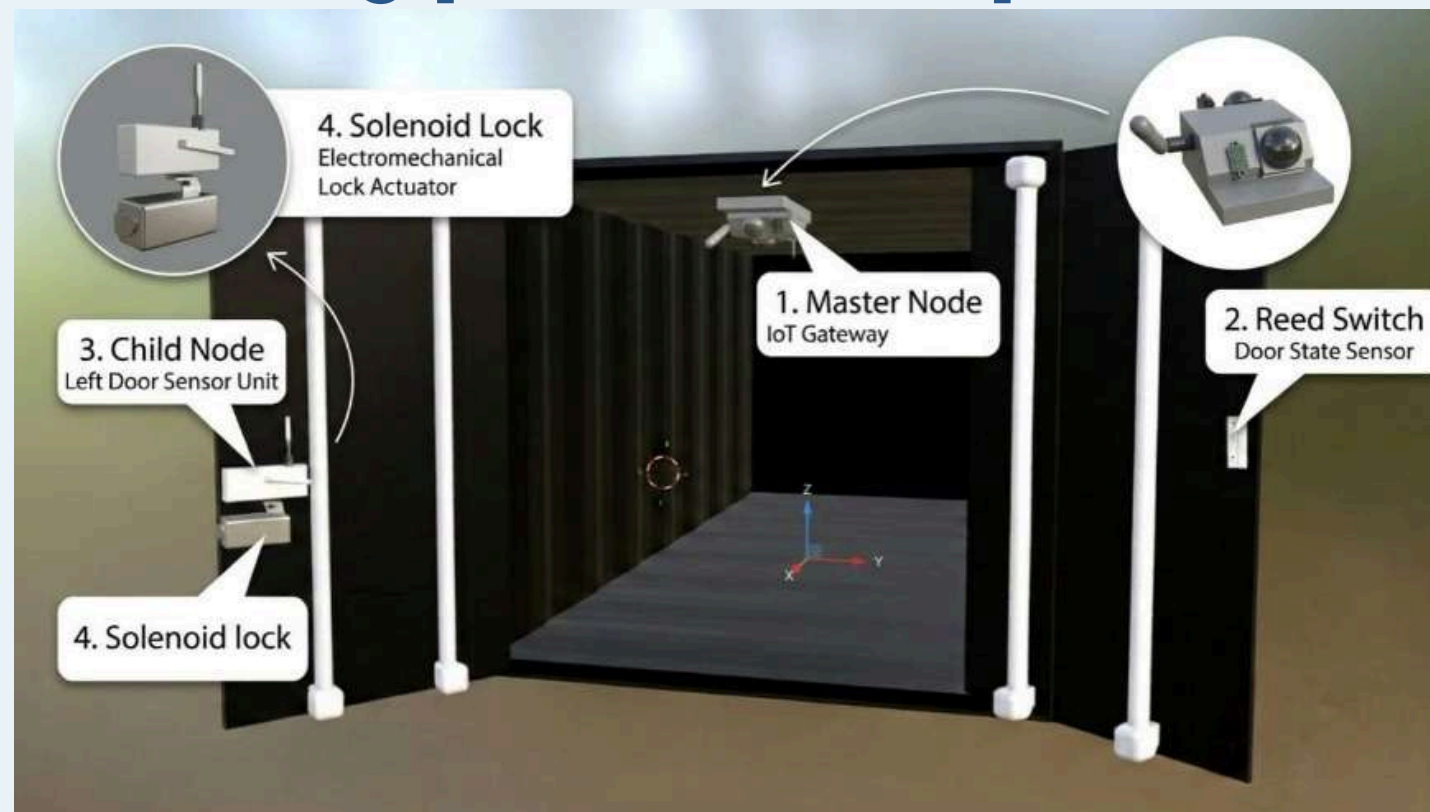
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System Architecture Diagram



Prototype Development



A prototype of the smart cargo container security system was developed using a dual-node architecture consisting of a **Master Node** and a **Child Node**. The **Child Node** is installed on the container door and includes an **ESP32 controller**, **solenoid lock**, **reed switch**, and **accelerometer** to monitor door status and lock tampering. The **Master Node** is placed inside the container and uses sensors such as **microwave radar**, **PIR**, and **VL53L5CX time-of-flight sensor** to monitor human movement and cargo displacement. The system processes sensor data using ESP32 and activates monitoring when suspicious activity is detected, improving cargo security.



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Hardware Components

Master Node (Main Monitoring Unit)

- **ESP32 Microcontroller** – System controller
- **Microwave Sensor (RCWL-0516)** – Motion detection
- **PIR Sensor** – Human presence detection
- **VL53L5CX ToF Sensor** – Cargo movement detection
- **ADXL345 Accelerometer** – Vibration / tilt detection
- **GPS Module (NEO-6M)** – Location tracking
- **DS3231 RTC Module** – Time stamping
- **18650 Battery + Voltage Regulator** – Power supply

Child Node (Smart Lock Unit)

- **ESP32 Microcontroller** – Lock monitoring controller
- **12V Solenoid Lock** – Electronic door lock
- **Magnetic Reed Switch** – Door status detection
- **ADXL345 Accelerometer** – Lock tampering detection



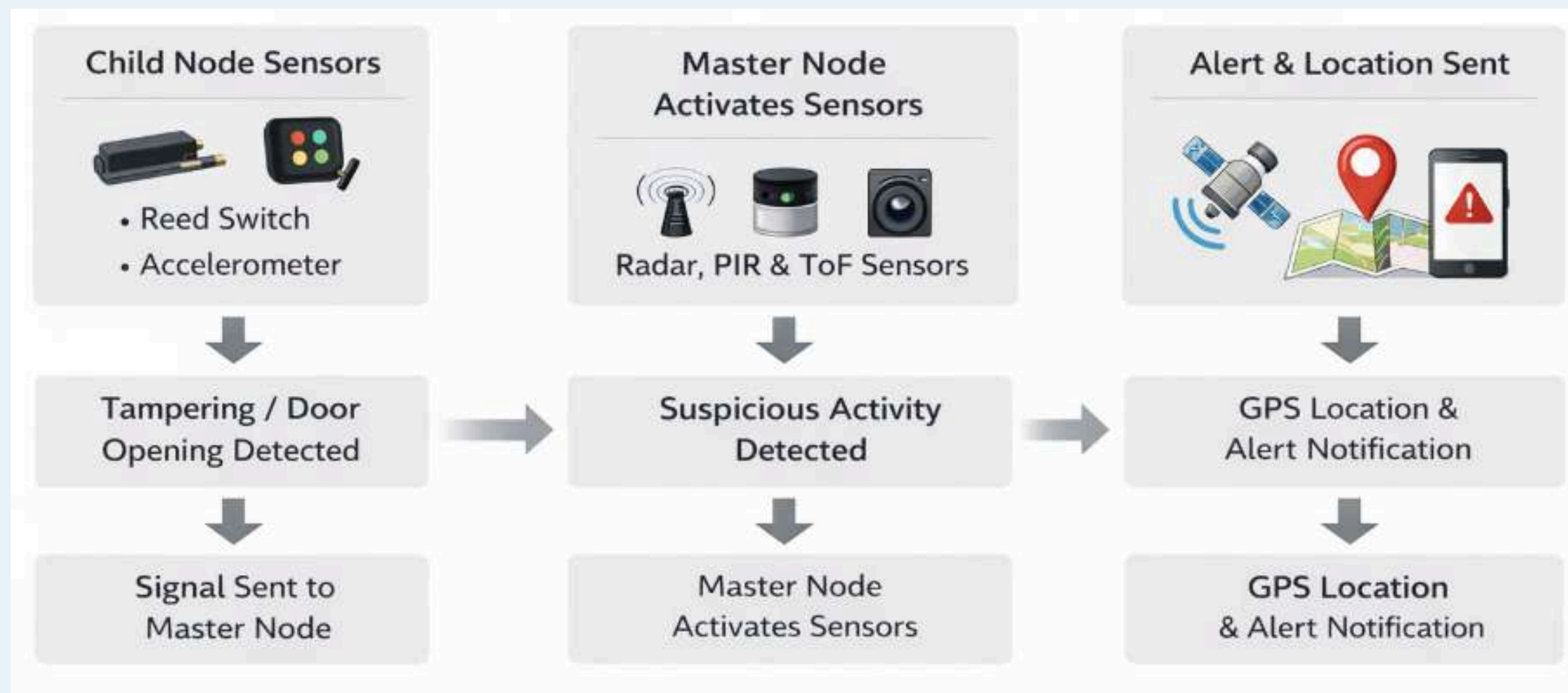
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System Workflow

How the cargo security system operates using Child Node and Master Node



The **Child Node** detects door tampering and sends a trigger signal to the Master Node.

The **Master Node activates sensors** to detect intrusion and **sends alert notifications with location data.**



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Findings

1. IoT-Based Intelligent Cargo Monitoring System. International Journal of Recent Progress in Research (IJRPR)- <https://ijrpr.com/uploads/V6ISSUE5/IJRPR45895.pdf>

2. Enhancing Safety in Cargo Handling Units Using Thermal and Motion IoT Sensors. International Journal For Multidisciplinary Research (IJFMR)- <https://www.ijfmr.com/research-paper.php?id=>

3.TL300 GPS-Enabled Smart Electronic Lock for Container Security-
<https://www.tracalogic.com/tl300>

4.Secure Container and Asset Monitoring System-<https://cypod.com/products/cylock>



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Conclusion

The **proposed IoT-based smart cargo container security system** **successfully detects** unauthorized access, cargo displacement, and lock tampering using multiple sensors. The integration of ESP32, motion sensors, and distance sensors enables **real-time monitoring of container activity**. The system provides an efficient and low-cost solution to improve cargo security during transportation. **Future improvements** can include **AI-based anomaly detection** and **geofencing** for enhanced monitoring.